

Multiple Choice

1. **Answer:** A

Options: A) Cu; B) Gd; C) Eu; D) Am

Note: Correct option: A - Cu

2. **Answer:** B

Options: A) NH³; B) NH²⁻; C) NH⁴⁺; D) N₂H₄

Note: Correct option: B - NH²⁻

3. **Answer:** D

Options: A) racemic mixture; B) single pure enantiomer; C) mixture of two diastereomers in equal amounts; D) meso compound

Note: Correct option: D - meso compound

4. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: Correct option: B - 42.58 MHz

5. **Answer:** D

Options: A) electron distribution; B) dipole moment; C) force constant; D) reduced mass

Note: Correct option: D - reduced mass

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'trigonal pyramidal'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The MnO 4- : I- ratio is 6:5.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Neither position nor momentum can be known exactly.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key